

CS620 | FINALTERM

Final Term MCQ Bank

300 questions | Original option sequence | Answer key

Inside this edition

All 300 questions from the uploaded CS620 files, merged in their original order (1-300). Question wording, available options and sequence are preserved word-by-word. Questions marked ★ MOST IMPORTANT come from past papers; questions marked REPEATED appeared in more than one source file.

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Answers independently verified against the CS620 Modeling & Simulation handouts and web sources; 14 source-marked answers were corrected during verification and 4 unanswerable questions were excluded.

COMPLETE QUESTION BANK

CS620 - Finalterm MCQs

300 questions • Same wording, options and sequence as the uploaded files.

QUESTION 001 A/An _____ topology is one in which all the edges are connected to another edge in a regular manner.

- A. Bounded
- B. Toroidal
- C. Non-Bounded
- D. Infinite Plane

CORRECT ANSWER B. Toroidal

QUESTION 002 In Von Neumann Neighborhood how many neighbors are there for each cell of square lattice.

REPEATED IN 2 FILES

- A. Three
- B. Two
- C. Five
- D. Four

CORRECT ANSWER D. Four

QUESTION 003 In Moore Neighborhood how many neighbors are there for each cell of square lattice.

REPEATED IN 2 FILES

- A. Six
- B. Four
- C. Two
- D. Eight

CORRECT ANSWER D. Eight

QUESTION 004 A network in which each node is randomly connected to other individual node is called as _____.

- A. Lattice Network
- B. Random Network
- C. Small-world Network
- D. Scale-free Network

CORRECT ANSWER B. Random Network

QUESTION 005 Small-world networks are made up of dense clusters of highly interconnected nodes joined by ____.

- A. No links
- B. A few long-distance links
- C. Many short-distance links
- D. Random links

CORRECT ANSWER B. A few long-distance links

QUESTION 006 The average of all the pairwise distances between nodes in a network is called ____.

- A. Clustering Coefficient
- B. Average Path Length
- C. Network Density
- D. Degree Centrality

CORRECT ANSWER B. Average Path Length

QUESTION 007 The clustering coefficient of a network is the average fraction of a node's immediate neighbors who are also ____.

- A. Far from the node
- B. Neighbors of the node's other neighbors
- C. Connected to random nodes
- D. Isolated from the network

CORRECT ANSWER B. Neighbors of the node's other neighbors

QUESTION 008 In a bounded topology, agents are ____.

- A. Allowed to move freely in any direction
- B. Not allowed to move beyond the edges of the world
- C. Allowed to wrap around the edges
- D. Allowed to expand the world

CORRECT ANSWER B. Not allowed to move beyond the edges of the world

QUESTION 009 Which topology allows agents to reappear on the opposite side when they move off the world?

- A. Bounded Topology
- B. Infinite-plane Topology
- C. Toroidal Topology
- D. Network Topology

CORRECT ANSWER C. Toroidal Topology

QUESTION 010 Which topology has no bounds and agents can keep moving forever in any direction?

- A. Bounded Topology
- B. Infinite-plane Topology
- C. Toroidal Topology
- D. Lattice Topology

CORRECT ANSWER B. Infinite-plane Topology

QUESTION 011 Which of the following is NOT a standard network topology used in ABM?

- A. Random Network
- B. Scale-free Network
- C. Small-world Network
- D. Toroidal Network

CORRECT ANSWER D. Toroidal Network

QUESTION 012 Scale-free networks have the property that ____.

- A. All nodes have the same number of connections
- B. Any subnetwork has the same properties as the global network
- C. Nodes are randomly connected
- D. Nodes form dense clusters

CORRECT ANSWER B. Any subnetwork has the same properties as the global network

QUESTION 013 In scale-free networks, a common way to create them is by ____.

- A. Randomly connecting all nodes
- B. Preferential attachment
- C. Removing existing nodes
- D. Creating dense clusters

CORRECT ANSWER B. Preferential attachment

QUESTION 014 Which network topology is also sometimes called hub-and-spoke?

- A. Random Network
- B. Lattice Network
- C. Scale-free Network
- D. Small-world Network

CORRECT ANSWER C. Scale-free Network

QUESTION 015 3D environments enable model developers to explore complex systems which are ____.

- A. Only two-dimensional
- B. Irreducibly bound up with a third dimension
- C. Only network-based
- D. Only lattice-based

CORRECT ANSWER B. Irreducibly bound up with a third dimension

QUESTION 016 In 3D NetLogo, the orientation of an agent is described by heading, pitch, and ____.

- A. Yaw
- B. Roll
- C. Tilt
- D. Angle

CORRECT ANSWER B. Roll

QUESTION 017 Geographic Information Systems (GIS) are environments that record large amounts of data related to ____.

- A. Network connections
- B. Physical locations in the world
- C. Agent properties
- D. Random numbers

CORRECT ANSWER B. Physical locations in the world

QUESTION 018 GIS can serve as an ____ for an ABM to operate in.

- A. Agent type
- B. Interaction topology
- C. Data structure
- D. Scheduling mechanism

CORRECT ANSWER B. Interaction topology

QUESTION 019 There are five basic classes of interactions in ABMs. Which of the following is NOT one of them?

- A. Agent-self
- B. Environment-self
- C. Agent-Agent
- D. Observer-Agent

CORRECT ANSWER D. Observer-Agent

QUESTION 020 Birth events are a classic type of ____ interaction in ABMs.

- A. Agent-agent
- B. Agent-environment
- C. Agent-self
- D. Environment-self

CORRECT ANSWER C. Agent-self

QUESTION 021 When grass regrows in a model, this is an example of ____ interaction.

- A. Agent-self
- B. Agent-environment
- C. Environment-self
- D. Agent-agent

CORRECT ANSWER C. Environment-self

QUESTION 022 When wolves consume sheep in a predator-prey model, this is an example of ____ interaction.

- A. Agent-self
- B. Agent-environment
- C. Environment-self
- D. Agent-agent

CORRECT ANSWER D. Agent-agent

QUESTION 023 Communication between agents is an example of ____ interaction.

- A. Agent-self
- B. Agent-environment
- C. Environment-environment
- D. Agent-agent

CORRECT ANSWER D. Agent-agent

QUESTION 024 Diffusion of pheromone in the Ants model is an example of ____ interaction.

- A. Agent-self
- B. Agent-environment
- C. Environment-environment
- D. Agent-agent

CORRECT ANSWER C. Environment-environment

QUESTION 025 When an ant examines the environment to look for food, this is an example of ____ interaction.

- A. Agent-self
- B. Agent-environment
- C. Environment-self
- D. Agent-agent

CORRECT ANSWER B. Agent-environment

QUESTION 026 The observer in an ABM is a ____ agent.

- A. Low-level
- B. High-level
- C. Mobile
- D. Stationary

CORRECT ANSWER B. High-level

QUESTION 027 The observer issues commands to ____.

- A. Only the environment
- B. Agents and the environment
- C. Only the agents
- D. Other observers

CORRECT ANSWER B. Agents and the environment

QUESTION 028 Which of the following is NOT a standard way to interface with an ABM?

- A. Buttons
- B. Sliders
- C. Switches
- D. Databases

CORRECT ANSWER D. Databases

QUESTION 029 Sliders in NetLogo enable the user to select a value from ____.

- A. A drop-down menu
- B. A range of numerical values
- C. A list of options
- D. A text input field

CORRECT ANSWER B. A range of numerical values

QUESTION 030 Switches in NetLogo are accessed as global variables that are ____.

- A. Numeric variables
- B. Boolean variables
- C. String variables
- D. List variables

CORRECT ANSWER B. Boolean variables

QUESTION 031 Choosers in NetLogo enable the user to select from ____.

- A. A range of numerical values
- B. A predefined drop-down menu
- C. A text input field
- D. A slider

CORRECT ANSWER B. A predefined drop-down menu

QUESTION 032 Monitors in NetLogo display ____.

- A. The value of a global variable or calculation
- B. A 2D graph
- C. A text description
- D. A list of agents

CORRECT ANSWER A. The value of a global variable or calculation

QUESTION 033 Plots in NetLogo provide ____.

- A. A text output
- B. A 2D graph
- C. A 3D visualization
- D. A monitor value

CORRECT ANSWER B. A 2D graph

QUESTION 034 Which of the following is a guideline for designing visualization of an ABM?

- A. Complicate the visualization
- B. Simplify the visualization
- C. Hide the components
- D. Avoid emphasizing the main point

CORRECT ANSWER B. Simplify the visualization

QUESTION 035 Which of the following is NOT a guideline for designing visualization of an ABM?

- A. Simplify
- B. Explain
- C. Emphasize
- D. Complicate

CORRECT ANSWER D. Complicate

QUESTION 036 Interactive running of a model means ____.

- A. Running the model with a script
- B. The user can interact as the model is running
- C. Running the model without any user input
- D. Running the model only once

CORRECT ANSWER B. The user can interact as the model is running

QUESTION 037 Batch running of a model involves ____.

- A. The user controlling the model directly
- B. Writing a script to run the model many times
- C. Running the model only once
- D. Running the model without any parameters

CORRECT ANSWER B. Writing a script to run the model many times

QUESTION 038 The SETUP procedure in a NetLogo model is usually used for ____.

- A. The main loop of the model
- B. Initialization of the model
- C. Updating the user interface
- D. Running the model multiple times

CORRECT ANSWER B. Initialization of the model

QUESTION 039 The GO procedure in a NetLogo model describes ____.

- A. What happens in one time unit of the model
- B. The initialization of the model
- C. The setup of the user interface
- D. The batch running of the model

CORRECT ANSWER A. What happens in one time unit of the model

QUESTION 040 If a model uses an asynchronous update schedule, this means that ____.

- A. All agents update simultaneously
- B. When agents change their state, that state is immediately seen by other agents
- C. Agents cannot change their state
- D. The model cannot run

CORRECT ANSWER B. When agents change their state, that state is immediately seen by other agents

QUESTION 041 In a synchronous update schedule, ____.

- A. Changes made to an agent are immediately seen by other agents
- B. Changes made to an agent are not seen by other agents until the next clock tick
- C. Agents cannot change their state
- D. The model runs faster

CORRECT ANSWER B. Changes made to an agent are not seen by other agents until the next clock tick

QUESTION 042 Sequential actions in ABM involve ____.

- A. All agents acting independently
- B. Only one agent acting at a time
- C. Agents acting in parallel
- D. Agents acting simultaneously

CORRECT ANSWER B. Only one agent acting at a time

QUESTION 043 Parallel actions in ABM involve ____.

- A. Only one agent acting at a time
- B. All agents acting independently
- C. Agents acting sequentially
- D. Agents cannot act in parallel

CORRECT ANSWER B. All agents acting independently

QUESTION 044 Simulated concurrency uses one processor to ____.

- A. Act as a single agent
- B. Simulate many agents acting in parallel
- C. Run the model only once
- D. Initialize the model

CORRECT ANSWER B. Simulate many agents acting in parallel

QUESTION 045 Descriptive statistics provide numerical measures that ____.

- A. Summarize a large data set
- B. Make the data more complex
- C. Hide the data
- D. Increase the data size

CORRECT ANSWER A. Summarize a large data set

QUESTION 046 In the Spread of Disease model, as population density increases, the time to full infection ____.

- A. Dramatically increases
- B. Dramatically decreases
- C. Remains the same
- D. Becomes unpredictable

CORRECT ANSWER B. Dramatically decreases

QUESTION 047 BehaviorSpace in NetLogo is a tool for ____.

- A. Creating agents
- B. Running the model multiple times with different settings
- C. Visualizing the model
- D. Debugging the model

CORRECT ANSWER B. Running the model multiple times with different settings

QUESTION 048 A chaotic system is a deterministic system that is ____.

- A. Easy to predict
- B. Difficult to predict
- C. Always stable
- D. Never changes

CORRECT ANSWER B. Difficult to predict

QUESTION 049 The term "chaos" in science has a precise mathematical meaning and refers to ____.

- A. Disorder or confusion
- B. A deterministic system that is difficult to predict
- C. A random system
- D. A stable system

CORRECT ANSWER B. A deterministic system that is difficult to predict

QUESTION 050 A deterministic system is defined as one whose state at one time ____.

- A. Completely determines its state for all future times
- B. Has no effect on future states
- C. Is random
- D. Is unpredictable

CORRECT ANSWER A. Completely determines its state for all future times

QUESTION 051 The butterfly effect is associated with ____.

- A. Linear systems
- B. Chaotic systems
- C. Stable systems
- D. Random systems

CORRECT ANSWER B. Chaotic systems

QUESTION 052 A complex adaptive system (CAS) is a system that ____.

- A. Never changes its behavior
- B. Changes its behavior in response to its environment
- C. Is always stable
- D. Has no interactions

CORRECT ANSWER B. Changes its behavior in response to its environment

QUESTION 053 The adaptive change that occurs in a CAS is often relevant to ____.

- A. Achieving a goal or objective
- B. Creating chaos
- C. Reducing complexity
- D. Eliminating agents

CORRECT ANSWER A. Achieving a goal or objective

QUESTION 054 The systems approach stands on the assumption that breaking down a complex concept into simple units helps in ____.

- A. Making it more complex
- B. Better understanding of the complexity
- C. Eliminating the system
- D. Creating chaos

CORRECT ANSWER B. Better understanding of the complexity

QUESTION 055 Ludwig von Bertalanffy first proposed the 'General System Theory' as _____.

- A. A theory of chaos
- B. A theory of systems
- C. A theory of randomness
- D. A theory of agents

CORRECT ANSWER B. A theory of systems

QUESTION 056 The CABC framework has how many levels for Agent-based modeling?

- A. Two
- B. Three
- C. Four
- D. Five

CORRECT ANSWER B. Three

QUESTION 057 Which of the following is a level of the CABC framework?

- A. Exploratory Agent-based modeling (EABM)
- B. Descriptive Agent-based Modeling (DREAM)
- C. Validated Agent-based Modeling (VABM)
- D. All of the above

CORRECT ANSWER D. All of the above

QUESTION 058 DREAM stands for _____.

- A. Descriptive Agent-based Modeling
- B. Dynamic Agent-based Modeling
- C. Distributed Agent-based Modeling
- D. Deterministic Agent-based Modeling

CORRECT ANSWER A. Descriptive Agent-based Modeling

QUESTION 059 VOMAS stands for _____.

- A. Virtual Overlay Multiagent System
- B. Validated Overlay Multiagent System
- C. Virtual Operating Multiagent System
- D. Validated Operating Multiagent System

CORRECT ANSWER A. Virtual Overlay Multiagent System

QUESTION 060 A graph and a network are ____.

- A. Completely different entities
- B. The same entity with different terms in different domains
- C. Always directed
- D. Always undirected

CORRECT ANSWER B. The same entity with different terms in different domains

QUESTION 061 A network can also be referred to as a ____.

- A. Weighted graph
- B. Unweighted graph
- C. Directed graph
- D. Undirected graph

CORRECT ANSWER A. Weighted graph

QUESTION 062 Which of the following is a benefit of complex network-based modeling?

- A. Very well-defined
- B. Lot of free tools
- C. Well-defined mathematical models
- D. All of the above

CORRECT ANSWER D. All of the above

QUESTION 063 Which of the following is a limitation of complex network-based modeling?

- A. Well-defined mathematical models
- B. Difficulty in graph mining
- C. Lot of free tools
- D. Very well-defined

CORRECT ANSWER B. Difficulty in graph mining

QUESTION 064 In Bernoulli distribution, each trial has ____.

- A. Multiple possible outcomes
- B. One out of two possible outcomes (success or failure)
- C. An infinite number of outcomes
- D. No outcomes

CORRECT ANSWER B. One out of two possible outcomes (success or failure)

QUESTION 065 The expected value of a Bernoulli random variable X is ____.

- A. p
- B. $1-p$
- C. $p(1-p)$
- D. 0

CORRECT ANSWER A. p

QUESTION 066 The variance of a Bernoulli random variable X is ____.

- A. p
- B. $1-p$
- C. $p(1-p)$
- D. 0

CORRECT ANSWER C. $p(1-p)$

QUESTION 067 Binomial distribution is the discrete probability distribution with parameters n and p of the number of successes in ____.

- A. A single trial
- B. A sequence of n independent experiments
- C. An infinite sequence
- D. A random sequence

CORRECT ANSWER B. A sequence of n independent experiments

QUESTION 068 The mean of a binomial distribution is ____.

- A. np
- B. $np(1-p)$
- C. p
- D. $1-p$

CORRECT ANSWER A. np

QUESTION 069 The variance of a binomial distribution is ____.

- A. np
- B. $np(1-p)$
- C. p
- D. $1-p$

CORRECT ANSWER B. $np(1-p)$

QUESTION 070 Poisson distribution gives the probability of independent events occurring in ____.

- A. A fixed interval of time
- B. An infinite interval of time
- C. A random interval of time
- D. No interval of time

CORRECT ANSWER A. A fixed interval of time

QUESTION 071 An important property of Poisson distribution is that ____.

- A. Mean is greater than variance
- B. Mean is less than variance
- C. Mean and variance are equal
- D. Mean and variance are not related

CORRECT ANSWER C. Mean and variance are equal

QUESTION 072 The uniform distribution is also known as ____.

- A. Rectangular distribution
- B. Triangular distribution
- C. Exponential distribution
- D. Normal distribution

CORRECT ANSWER A. Rectangular distribution

QUESTION 073 The probability density function for a uniform distribution always takes a random number on ____.

- A. $[0, 1]$
- B. $[0, \infty]$
- C. $[-\infty, \infty]$
- D. $[1, 2]$

CORRECT ANSWER A. $[0, 1]$

QUESTION 074 The Exponential Distribution is a ____ valued probability distribution.

- A. Discrete
- B. Continuous
- C. Binary
- D. Integer

CORRECT ANSWER B. Continuous

QUESTION 075 In Exponential Distribution, λ is called the ____.

- A. Rate parameter
- B. Shape parameter
- C. Scale parameter
- D. Location parameter

CORRECT ANSWER A. Rate parameter

QUESTION 076 The inverse transform technique is used when ____.

- A. The cdf is very complex
- B. The form of cdf is so simple that its F^{-1} can be computed easily
- C. The distribution is discrete
- D. The distribution is unknown

CORRECT ANSWER B. The form of cdf is so simple that its F^{-1} can be computed easily

QUESTION 077 A triangular distribution is a continuous probability distribution with a probability density function shaped like a ____.

- A. Rectangle
- B. Triangle
- C. Circle
- D. Bell

CORRECT ANSWER B. Triangle

QUESTION 078 A triangular distribution is defined by ____.

- A. Minimum value a, maximum value b, peak value c
- B. Mean and variance
- C. Rate parameter only
- D. Shape parameter only

CORRECT ANSWER A. Minimum value a, maximum value b, peak value c

QUESTION 079 A Customer in a queueing system is ____.

- A. Anything that visits a facility and requires some service
- B. Only a person
- C. Only a machine
- D. Only a computer

CORRECT ANSWER A. Anything that visits a facility and requires some service

QUESTION 080 The calling population is the ____.

- A. Population of potential customers
- B. Population of servers
- C. Population of machines
- D. Population of computers

CORRECT ANSWER A. Population of potential customers

QUESTION 081 A calling population can be ____.

- A. Finite or infinite
- B. Only finite
- C. Only infinite
- D. Neither finite nor infinite

CORRECT ANSWER A. Finite or infinite

QUESTION 082 The arrival rate is the main difference between ____.

- A. Finite and infinite population models
- B. Single server and multiple server models
- C. FIFO and LIFO models
- D. M/M/1 and M/M/c models

CORRECT ANSWER A. Finite and infinite population models

QUESTION 083 Every queueing system has some ____.

- A. Capacity
- B. Server
- C. Customer
- D. Queue

CORRECT ANSWER A. Capacity

QUESTION 084 If the entering system finds the capacity full, it will ____.

- A. Wait indefinitely
- B. Return to the calling population
- C. Create more capacity
- D. Remove other customers

CORRECT ANSWER B. Return to the calling population

QUESTION 085 Poisson arrival process is the most important for ____.

- A. Scheduled arrivals
- B. Random arrivals
- C. Constant arrivals
- D. Infinite arrivals

CORRECT ANSWER B. Random arrivals

QUESTION 086 In Poisson arrival process, the number of arrivals in a time interval of length t has ____.

- A. Exponential distribution with mean λt
- B. Poisson distribution with mean λt
- C. Normal distribution with mean λt
- D. Uniform distribution with mean λt

CORRECT ANSWER B. Poisson distribution with mean λt

QUESTION 087 Queue behavior refers to ____.

- A. The actions of the customer while waiting in the queue
- B. The logical ordering of customers in the queue
- C. The service time distribution
- D. The number of servers

CORRECT ANSWER A. The actions of the customer while waiting in the queue

QUESTION 088 Queue discipline refers to ____.

- A. The actions of the customer while waiting in the queue
- B. The logical ordering of the customers in a queue
- C. The service time distribution
- D. The number of servers

CORRECT ANSWER B. The logical ordering of the customers in a queue

QUESTION 089 Which of the following is a common queue discipline?

- A. FIFO
- B. LIFO
- C. SIRO
- D. All of the above

CORRECT ANSWER D. All of the above

QUESTION 090 In Kendall's queueing notation, "A" represents ____.

- A. The service-time distribution
- B. The inter-arrival-time distribution
- C. The number of parallel servers
- D. The system capacity

CORRECT ANSWER B. The inter-arrival-time distribution

QUESTION 091 In Kendall's queueing notation, "B" represents ____.

- A. The inter-arrival-time distribution
- B. The service-time distribution
- C. The number of parallel servers
- D. The system capacity

CORRECT ANSWER B. The service-time distribution

QUESTION 092 In Kendall's queueing notation, "M" represents ____.

- A. Exponential or Markov
- B. Constant or deterministic
- C. Arbitrary or general
- D. Erlang

CORRECT ANSWER A. Exponential or Markov

QUESTION 093 In Kendall's queueing notation, "D" represents ____.

- A. Exponential or Markov
- B. Constant or deterministic
- C. Arbitrary or general
- D. Erlang

CORRECT ANSWER B. Constant or deterministic

QUESTION 094 The two important statistical properties of random numbers are ____.

- A. Uniformity and independence
- B. Uniformity and dependence
- C. Randomness and independence
- D. Randomness and dependence

CORRECT ANSWER A. Uniformity and independence

QUESTION 095 A random number should be drawn from ____.

- A. Continuous uniform distribution between 0 and 1
- B. Discrete uniform distribution between 0 and 1
- C. Normal distribution between 0 and 1
- D. Exponential distribution between 0 and 1

CORRECT ANSWER A. Continuous uniform distribution between 0 and 1

QUESTION 096 The expected value of a random number R_i is ____.

- A. 0
- B. 1
- C. $1/2$
- D. $1/3$

CORRECT ANSWER C. $1/2$

QUESTION 097 The variance of a random number R_i is ____.

- A. $1/2$
- B. $1/4$
- C. $1/12$
- D. $1/3$

CORRECT ANSWER C. $1/12$

QUESTION 098 The Linear Congruential Method is a widely used technique that ____.

- A. Yields sequences with shorter periods
- B. Yields sequences with longer periods
- C. Produces truly random numbers
- D. Cannot be replicated

CORRECT ANSWER B. Yields sequences with longer periods

QUESTION 099 In the Linear Congruential Method, when $c = 0$, it is known as ____.

- A. Mixed congruential method
- B. Multiplicative congruential method
- C. Additive congruential method
- D. Exponential congruential method

CORRECT ANSWER B. Multiplicative congruential method

QUESTION 100 In the Linear Congruential Method, when $c \neq 0$, it is known as ____.

- A. Mixed congruential method
- B. Multiplicative congruential method
- C. Additive congruential method
- D. Exponential congruential method

CORRECT ANSWER A. Mixed congruential method

QUESTION 101 Pseudo-random numbers are numbers generated that ____.

- A. Are truly random
- B. Simulate the properties of uniform distribution and independence
- C. Cannot be replicated
- D. Have no pattern

CORRECT ANSWER B. Simulate the properties of uniform distribution and independence

QUESTION 102 Tests for random numbers are divided into two categories: ____.

- A. Uniformity and independence
- B. Uniformity and dependence
- C. Randomness and independence
- D. Randomness and dependence

CORRECT ANSWER A. Uniformity and independence

QUESTION 103 A level of significance α is the probability of ____.

- A. Accepting the null hypothesis
- B. Rejecting the null hypothesis
- C. Accepting the alternative hypothesis
- D. Rejecting the alternative hypothesis

CORRECT ANSWER B. Rejecting the null hypothesis

QUESTION 104 A random variate is a generated variable using ____.

- A. Truly random numbers
- B. Uniformly distributed pseudorandom numbers
- C. Exponentially distributed pseudorandom numbers
- D. Normally distributed pseudorandom numbers

CORRECT ANSWER B. Uniformly distributed pseudorandom numbers

QUESTION 105 Which of the following is a technique for generating random variates?

- A. Inverse-transform technique
- B. Acceptance-rejection technique
- C. Both A and B
- D. Neither A nor B

CORRECT ANSWER C. Both A and B

QUESTION 106 The inverse-transform technique can be used to sample from which distribution?

- A. Exponential
- B. Uniform
- C. Weibull
- D. All of the above

CORRECT ANSWER D. All of the above

QUESTION 107 The basic principle of inverse-transform technique is to find ____.

- A. The inverse function of F , F^{-1}
- B. The derivative of F
- C. The integral of F
- D. The limit of F

CORRECT ANSWER A. The inverse function of F , F^{-1}

QUESTION 108 The acceptance-rejection technique is used to ____.

- A. Generate random variates
- B. Accept all random numbers
- C. Reject all random numbers
- D. Generate only uniform random numbers

CORRECT ANSWER A. Generate random variates

QUESTION 109 The efficiency of an acceptance-rejection technique rests on ____.

- A. Increasing the total number of rejections
- B. Reducing the total number of rejections
- C. Accepting all random numbers
- D. Rejecting all random numbers

CORRECT ANSWER B. Reducing the total number of rejections

QUESTION 110 Direct transformation is used for generating which distribution?

- A. Uniform distribution
- B. Normal and lognormal distributions
- C. Exponential distribution
- D. Triangular distribution

CORRECT ANSWER B. Normal and lognormal distributions

QUESTION 111 The Box and Muller method is used for generating ____.

- A. Uniform random numbers
- B. Normal random variates
- C. Exponential random variates
- D. Triangular random variates

CORRECT ANSWER B. Normal random variates

QUESTION 112 The probability distribution of a sum of two or more independent random variables is called ____.

- A. Direct transformation
- B. Convolution method
- C. Acceptance-rejection method
- D. Inverse-transform method

CORRECT ANSWER B. Convolution method

QUESTION 113 Histograms are also known as ____.

- A. Frequency distribution
- B. Probability distribution
- C. Cumulative distribution
- D. Density function

CORRECT ANSWER A. Frequency distribution

QUESTION 114 The selection of a family of distributions is based on ____.

- A. The context of what is being investigated
- B. The shape of the histogram
- C. Both A and B
- D. Neither A nor B

CORRECT ANSWER C. Both A and B

QUESTION 115 The Quantile-Quantile plot is a graphical method for ____.

- A. Estimating parameters
- B. Comparing two probability distributions by plotting their quantiles against each other
- C. Creating histograms
- D. Calculating the mean

CORRECT ANSWER B. Comparing two probability distributions by plotting their quantiles against each other

QUESTION 116 After selecting the family of distribution, the next step is ____.

- A. Data collection
- B. Parameter estimation
- C. Model verification
- D. Model validation

CORRECT ANSWER B. Parameter estimation

QUESTION 117 The sample mean is denoted by ____.

- A. $\bar{x}$
- B. s^2
- C. σ^2
- D. μ

CORRECT ANSWER A. $\bar{x}$

QUESTION 118 The sample variance is denoted by ____.

- A. $\bar{x}$
- B. s^2
- C. σ^2
- D. μ

CORRECT ANSWER B. s^2

QUESTION 119 ENIAC was the first computer system with ____.

- A. A good operating system
- B. No operating system
- C. A database system
- D. A network system

CORRECT ANSWER B. No operating system

QUESTION 120 RISC stands for ____.

- A. Reduced Instruction Set Computer
- B. Random Instruction Set Computer
- C. Regular Instruction Set Computer
- D. Reliable Instruction Set Computer

CORRECT ANSWER A. Reduced Instruction Set Computer

QUESTION 121 CISC stands for ____.

- A. Complex Instruction Set Computer
- B. Compound Instruction Set Computer
- C. Central Instruction Set Computer
- D. Comprehensive Instruction Set Computer

CORRECT ANSWER A. Complex Instruction Set Computer

QUESTION 122 ACID properties in database systems include ____.

- A. Atomicity, Consistency, Isolation, Durability
- B. Atomicity, Concurrency, Isolation, Durability
- C. Availability, Consistency, Isolation, Durability
- D. Atomicity, Consistency, Integrity, Durability

CORRECT ANSWER A. Atomicity, Consistency, Isolation, Durability

QUESTION 123 An operating system is the most privileged of ____ on the system.

- A. Hardware elements
- B. Software elements
- C. Firmware elements
- D. Database elements

CORRECT ANSWER B. Software elements

QUESTION 124 ARPANET was an early ____.

- A. Operating system
- B. Database system
- C. Computer network
- D. Programming language

CORRECT ANSWER C. Computer network

QUESTION 125 LAN stands for ____.

- A. Local Area Network
- B. Large Area Network
- C. Logical Area Network
- D. Linear Area Network

CORRECT ANSWER A. Local Area Network

QUESTION 126 Wireless technology for networks began in the 1970s with ____ as foundation.

- A. Ethernet
- B. ARPANET
- C. Aloha net
- D. Token ring

CORRECT ANSWER C. Aloha net

QUESTION 127 Performance evaluation is needed when ____.

- A. Selecting a specific computer architecture
- B. Selecting an operating system
- C. Selecting a database system
- D. All of the above

CORRECT ANSWER D. All of the above

QUESTION 128 A model provides a tool to ____.

- A. Define a system
- B. Study a system before its existence
- C. Both A and B
- D. Neither A nor B

CORRECT ANSWER C. Both A and B

QUESTION 129 Modeling is better if ____.

- A. Physical laws are available
- B. Pictorial representation is possible
- C. Input/output elements are manageable
- D. All of the above

CORRECT ANSWER D. All of the above

QUESTION 130 The goal of comparing two system designs is to obtain ____.

- A. Point estimates
- B. Interval estimates of the difference in mean performance
- C. Both A and B
- D. Neither A nor B

CORRECT ANSWER C. Both A and B

QUESTION 131 ____ enables the model user to select a value from a range of numerical values.

- A. Sliders
- B. Buttons
- C. Input Boxes
- D. Switches

CORRECT ANSWER A. Sliders

QUESTION 132 ____ networks are made up of dense clusters of highly interconnected nodes joined by a few long-distance links between them.

- A. Random Network
- B. Lattice Network
- C. Small-world Network
- D. Scale-free Network

CORRECT ANSWER C. Small-world Network

QUESTION 133 Agents that cannot move at the landscape are known as ____.

- A. Mobile Agents
- B. Model-based Agents
- C. Stationary Agents
- D. Connecting Agents

CORRECT ANSWER C. Stationary Agents

QUESTION 134 Model ____ is the implementation by one researchers of a conceptual model previously implemented by someone else.

- A. Verification
- B. Validation
- C. Replication
- D. Correctness

CORRECT ANSWER C. Replication

QUESTION 135 For different Models the granularity is ____.

- A. Multiple
- B. Equal
- C. Different
- D. Similar

CORRECT ANSWER C. Different

QUESTION 136 Which Statement is not true for observers

REPEATED IN 2 FILES

- A. It is a low-level agent
- B. The observer issues commands to agents and the environment
- C. It ensures that the model runs and proceeds according to the steps developed by the model author
- D. It is a high-level agent

CORRECT ANSWER A. It is a low-level agent

QUESTION 137 There are ____ keyways of doing model documentation.

- A. Two
- B. Three
- C. Four
- D. Five

CORRECT ANSWER A. Two

QUESTION 138 Agent-based methods may be particularly valuable in thesciences.

★ **MOST IMPORTANT - PAST PAPER** • **REPEATED IN 2 FILES**

- A. SOCIAL
- B. NONE
- C. BIOLOGICAL
- D. PHYSICAL

CORRECT ANSWER A. SOCIAL

QUESTION 139 Agent cognition is defined as how much _____ is added into the agents.

- A. Complexity
- B. Granularity
- C. Coherence
- D. Simplicity

CORRECT ANSWER A. Complexity

QUESTION 140 In agent-based modeling the environment consists of _____ surrounding the agents as they act and interact within the model.

- A. Habitats and decisions
- B. Conditions
- C. Conditions & habitats
- D. Conditions and decisions

CORRECT ANSWER C. Conditions & habitats

QUESTION 141 _____ enables the model user to select a value from a range of numerical values.

- A. Sliders
- B. Buttons
- C. Input Boxes
- D. Switches

CORRECT ANSWER A. Sliders

QUESTION 142 _____ networks are made up of dense clusters of highly interconnected nodes joined by a few long-distance links between them.

- A. Random Network
- B. Lattice Network
- C. Small-world Network
- D. Scale-free Network

CORRECT ANSWER C. Small-world Network

QUESTION 143 Communication is an example of _____ interaction.

- A. Agent - self
- B. Agent - Environment
- C. Environment - Environment
- D. Agent - Agent

CORRECT ANSWER D. Agent - Agent

QUESTION 144 Agents that cannot move at the landscape are known as ____.

- A. Mobile Agents
- B. Model-based Agents
- C. Stationary Agents
- D. Connecting Agents

CORRECT ANSWER C. Stationary Agents

QUESTION 145 Model ____ is the implementation by one researchers of a conceptual model previously implemented by someone else.

- A. Verification
- B. Validation
- C. Replication
- D. Correctness

CORRECT ANSWER C. Replication

QUESTION 146 A/An ____ topology is one in which all the edges are connected to another edge in a regular manner.

- A. Bounded
- B. Toroidal
- C. Non-Bounded
- D. Infinite Plane

CORRECT ANSWER B. Toroidal

QUESTION 147 A network in which each node is randomly connected to other individual node is called as ____.

- A. Lattice Network
- B. Random Network
- C. Small-world Network
- D. Scale-free Network

CORRECT ANSWER B. Random Network

QUESTION 148 For different Models the granularity is ____.

- A. Multiple
- B. Equal
- C. Different
- D. Similar

CORRECT ANSWER C. Different

QUESTION 149 There are _____ keyways of doing model documentation.

- A. Two
- B. Three
- C. Four
- D. Five

CORRECT ANSWER A. Two

QUESTION 150 Agent-based methods may be particularly valuable in thesciences.

- A. SOCIAL
- B. NONE
- C. BIOLOGICAL
- D. PHYSICAL

CORRECT ANSWER A. SOCIAL

QUESTION 151 Agent cognition is defined as how much _____ is added into the agents.

- A. Complexity
- B. Granularity
- C. Coherence
- D. Simplicity

CORRECT ANSWER A. Complexity

QUESTION 152 In agent-based modeling the environment consists of _____ surrounding the agents as they act and interact within the model.

- A. Habitats and decisions
- B. Conditions
- C. Conditions & habitats
- D. Conditions and decisions

CORRECT ANSWER C. Conditions & habitats

QUESTION 153 What are the two main types of spatial environments in agent-based models?

REPEATED IN 2 FILES

- A. Finite and Infinite Spaces
- B. Lattice and Network Spaces
- C. Discrete and Continuous Spaces
- D. Real and Hypothetical Spaces

CORRECT ANSWER C. Discrete and Continuous Spaces

QUESTION 154 In continuous spaces, what is the key mathematical property between any two points?

REPEATED IN 2 FILES

- A. The distance is always infinite.
- B. There exists another point between them.
- C. The points are disconnected.
- D. The distance is always zero.

CORRECT ANSWER B. There exists another point between them.

QUESTION 155 Which of the following is the most common representation of discrete spaces in agent-based models?

REPEATED IN 2 FILES

- A. Square lattices
- B. Hexagonal grids
- C. Random graphs
- D. Circular networks

CORRECT ANSWER A. Square lattices

QUESTION 156 What does a von Neumann neighborhood of radius 1 consist of?

REPEATED IN 2 FILES

- A. All points within a hexagonal shape.
- B. Four neighbors: up, down, left, and right.
- C. Eight neighbors in all diagonal directions.
- D. Only the closest point to the origin.

CORRECT ANSWER B. Four neighbors: up, down, left, and right.

QUESTION 157 Which topology allows an agent to reappear on the opposite edge of the lattice?

REPEATED IN 2 FILES

- A. Infinite plane topology
- B. Bounded topology
- C. Toroidal topology
- D. Moore topology

CORRECT ANSWER C. Toroidal topology

QUESTION 158 What type of network is characterized by dense clusters of nodes connected by a few long-distance links?

REPEATED IN 2 FILES

- A. Random network
- B. Scale-free network
- C. Small-world network
- D. Lattice network

CORRECT ANSWER C. Small-world network

QUESTION 159 In a small-world network, which metric measures the distance between any two nodes?

REPEATED IN 2 FILES

- A. Clustering coefficient
- B. Average path length
- C. Node centrality
- D. Degree distribution

CORRECT ANSWER B. Average path length

QUESTION 160 Which statistical measure is most commonly used to summarize agent-based model outputs?

REPEATED IN 2 FILES

- A. Median
- B. Standard Deviation
- C. Skewness
- D. Interquartile Range

CORRECT ANSWER B. Standard Deviation

QUESTION 161 What is a common method for verifying the correctness of a model?

REPEATED IN 2 FILES

- A. Using random inputs
- B. Comparing results with real-world data
- C. Applying unsupervised learning
- D. Ignoring validation steps

CORRECT ANSWER B. Comparing results with real-world data

QUESTION 162 Which of the following is an example of agent-agent interaction in an agent-based model?

REPEATED IN 2 FILES

- A. Grass regrowing on a patch
- B. A wolf consuming a sheep
- C. An agent moving across a grid
- D. A pheromone diffusing through the environment

CORRECT ANSWER B. A wolf consuming a sheep

QUESTION 163 What is the primary benefit of "randomized initialization" in ABMs?

REPEATED IN 2 FILES

- A. Ensures identical results for every simulation
- B. Reflects realistic variability in agent behavior
- C. Eliminates the need for parameter tuning
- D. Reduces computational complexity

CORRECT ANSWER B. Reflects realistic variability in agent behavior

QUESTION 164 Which method is commonly used to reduce the computational cost of ABMs?

REPEATED IN 2 FILES

- A. Increasing the agent population
- B. Parallelizing simulations across processors
- C. Avoiding random number generation
- D. Simplifying agent interaction rules

CORRECT ANSWER B. Parallelizing simulations across processors

QUESTION 165 What does the "ask" primitive in NetLogo allow agents to do?

REPEATED IN 2 FILES

- A. Communicate directly with the observer
- B. Execute commands specific to their type
- C. Randomly change their properties
- D. Alter global variables without restrictions

CORRECT ANSWER B. Execute commands specific to their type

QUESTION 166 In ABMs, what is a "rule-based approach"?

REPEATED IN 2 FILES

- A. Using machine learning to determine agent behavior
- B. Defining agent actions based on explicit conditions
- C. Allowing agents to behave randomly
- D. Preprogramming global outcomes for the model

CORRECT ANSWER B. Defining agent actions based on explicit conditions

QUESTION 167 Which tool in NetLogo is used for automated parameter testing?

REPEATED IN 2 FILES

- A. View screen
- B. BehaviorSpace
- C. Command Center
- D. Histogram widget

CORRECT ANSWER B. BehaviorSpace

QUESTION 168 What is the typical purpose of "thresholds" in agent decision-making?

REPEATED IN 2 FILES

- A. To limit the range of agent movement
- B. To trigger specific actions based on conditions
- C. To randomly select agent behaviors
- D. To simplify the interaction environment

CORRECT ANSWER B. To trigger specific actions based on conditions

QUESTION 169 Which network property measures how well-connected the neighbors of a node are?

REPEATED IN 2 FILES

- A. Clustering coefficient
- B. Degree centrality
- C. Betweenness centrality
- D. Average path length

CORRECT ANSWER A. Clustering coefficient

QUESTION 170 What does "agent memory" refer to in ABMs?

REPEATED IN 2 FILES

- A. The ability of agents to move within a fixed area
- B. The ability of agents to store and use past information
- C. The predefined rules all agents must follow
- D. The global variables shared among all agents

CORRECT ANSWER B. The ability of agents to store and use past information

QUESTION 171 What is a primary application of ABMs in ecology?

REPEATED IN 2 FILES

- A. Modeling urban development patterns
- B. Simulating species interactions in ecosystems
- C. Studying financial market dynamics
- D. Optimizing industrial processes

CORRECT ANSWER B. Simulating species interactions in ecosystems

QUESTION 172 Which type of visualization is best suited for observing agent interactions over time?

REPEATED IN 2 FILES

- A. Time-series plots
- B. Heatmaps
- C. 3D scatterplots
- D. Static network diagrams

CORRECT ANSWER A. Time-series plots

QUESTION 173 What is the primary difference between a synchronous and asynchronous update in ABMs?

REPEATED IN 2 FILES

- A. Asynchronous updates involve fixed time intervals.
- B. Synchronous updates occur simultaneously for all agents.
- C. Asynchronous updates allow agents to interact globally.
- D. Synchronous updates prevent agent interactions.

CORRECT ANSWER B. Synchronous updates occur simultaneously for all agents.

QUESTION 174 What is the role of the "breed" keyword in NetLogo?

REPEATED IN 2 FILES

- A. To specify agent-specific variables
- B. To create subgroups of agents
- C. To define environmental properties
- D. To randomize agent movements

CORRECT ANSWER B. To create subgroups of agents

QUESTION 175 Which process is essential for validating an ABM?

REPEATED IN 2 FILES

- A. Running multiple simulations without stopping
- B. Comparing simulation outcomes to observed data
- C. Eliminating randomness from the model
- D. Adding more rules to agents

CORRECT ANSWER B. Comparing simulation outcomes to observed data

QUESTION 176 What does "agent density" refer to in ABMs?

REPEATED IN 2 FILES

- A. The number of agents per unit area
- B. The frequency of agent interactions
- C. The average speed of agent movement
- D. The complexity of agent rules

CORRECT ANSWER A. The number of agents per unit area

QUESTION 177 In NetLogo, what is the purpose of "link" agents?

REPEATED IN 2 FILES

- A. To visualize networks between agents
- B. To define movement constraints for turtles
- C. To store global variables for patches
- D. To simplify agent properties

CORRECT ANSWER A. To visualize networks between agents

QUESTION 178 What type of schedule ensures that all agents update their states simultaneously?

REPEATED IN 2 FILES

- A. Asynchronous schedule
- B. Sequential schedule
- C. Synchronous schedule
- D. Randomized schedule

CORRECT ANSWER C. Synchronous schedule

QUESTION 179 Which model in NetLogo demonstrates agent-agent interactions like predation?

REPEATED IN 2 FILES

- A. Traffic Basic model
- B. Wolf-Sheep Predation model
- C. Ants model
- D. Disease Spread model

CORRECT ANSWER B. Wolf-Sheep Predation model

QUESTION 180 In which type of network are nodes connected without any specific pattern?

REPEATED IN 2 FILES

- A. Scale-free network
- B. Random network
- C. Small-world network
- D. Grid network

CORRECT ANSWER B. Random network

QUESTION 181 Which factor primarily influences the movement of agents in a GIS-based ABM?

REPEATED IN 2 FILES

- A. Random number generation
- B. Geographic constraints
- C. Social network connections
- D. Interaction rules

CORRECT ANSWER B. Geographic constraints

QUESTION 182 What is the primary purpose of sensitivity analysis in ABMs?

REPEATED IN 2 FILES

- A. To increase computational efficiency
- B. To test the impact of parameter changes on model outcomes
- C. To simplify the model design
- D. To verify model outputs against real-world data

CORRECT ANSWER B. To test the impact of parameter changes on model outcomes

QUESTION 183 Which NetLogo feature allows users to manually test commands on agents during runtime?

REPEATED IN 2 FILES

- A. Command Center
- B. Batch Runner
- C. BehaviorSpace
- D. Data Monitor

CORRECT ANSWER A. Command Center

QUESTION 184 What is a Moore neighborhood of radius 1 in a square lattice?

REPEATED IN 2 FILES

- A. Includes only diagonal neighbors.
- B. Includes neighbors up, down, left, and right.
- C. Includes all eight surrounding neighbors.
- D. Only includes immediate horizontal neighbors.

CORRECT ANSWER C. Includes all eight surrounding neighbors.

QUESTION 185 What is the key characteristic of an infinite-plane topology in ABMs?

REPEATED IN 2 FILES

- A. It loops edges of the grid to connect opposite sides.
- B. Agents can move indefinitely in any direction.
- C. It restricts agents to a bounded area.
- D. It requires all agents to remain static.

CORRECT ANSWER B. Agents can move indefinitely in any direction.

QUESTION 186 Which of the following is a significant drawback of random networks?

REPEATED IN 2 FILES

- A. Lack of predictable structure
- B. Excessive computational resources required
- C. Inability to model social interactions
- D. High clustering coefficients

CORRECT ANSWER A. Lack of predictable structure

QUESTION 187 What is the primary benefit of using BehaviorSpace in NetLogo?

REPEATED IN 2 FILES

- A. It allows for creating custom topologies.
- B. It automates multiple runs of a simulation with varying parameters.
- C. It visualizes agent paths in 3D environments.
- D. It performs real-time agent interactions.

CORRECT ANSWER B. It automates multiple runs of a simulation with varying parameters.

QUESTION 188 How do agents interact with their environment in the "Ants" model?

REPEATED IN 2 FILES

- A. By observing the environment and making pheromones.
- B. By forming hierarchies among themselves.
- C. By splitting into multiple sub-agents.
- D. By ignoring environmental changes.

CORRECT ANSWER A. By observing the environment and making pheromones.

QUESTION 189 What is the main difference between synchronous and asynchronous updates in ABMs?

REPEATED IN 2 FILES

- A. Synchronous updates only apply to spatial models.
- B. Asynchronous updates allow agents to update at different times.
- C. Synchronous updates are unpredictable.
- D. Asynchronous updates are computationally more expensive.

CORRECT ANSWER B. Asynchronous updates allow agents to update at different times.

QUESTION 190 What does "patches" refer to in NetLogo's 2D environments?

REPEATED IN 2 FILES

- A. Fixed agents located on a 3D plane
- B. Grid cells representing discrete spaces
- C. Dynamic agents with changing properties
- D. Randomly generated graphs

CORRECT ANSWER B. Grid cells representing discrete spaces

QUESTION 191 Which model component ensures reproducibility in agent-based simulations?

REPEATED IN 2 FILES

- A. Pseudo-random number generation
- B. Increasing agent density
- C. Batch experiments without recording
- D. Eliminating boundary conditions

CORRECT ANSWER A. Pseudo-random number generation

QUESTION 192 In ABMs, what does "diffusion" typically represent?

REPEATED IN 2 FILES

- A. The spread of resources or information through the environment
- B. The movement of agents into empty spaces
- C. The collapse of clusters in small-world networks
- D. A reduction in the number of active agents

CORRECT ANSWER A. The spread of resources or information through the environment

QUESTION 193 Which process ensures that agent behaviors align with intended rules in a model?

REPEATED IN 2 FILES

- A. Validation
- B. Debugging
- C. Initialization
- D. Verification

CORRECT ANSWER D. Verification

QUESTION 194 How does "clustering coefficient" affect network structure in small-world models?

REPEATED IN 2 FILES

- A. It determines the average path length.
- B. It measures the likelihood of node neighbors being interconnected.
- C. It defines the total number of edges in the network.
- D. It influences the randomness of the network.

CORRECT ANSWER B. It measures the likelihood of node neighbors being interconnected.

QUESTION 195 What is the role of "setup" procedures in NetLogo models?

REPEATED IN 2 FILES

- A. They execute model simulations.
- B. They initialize the environment and agents.
- C. They update agent states after each tick.
- D. They display output graphs.

CORRECT ANSWER B. They initialize the environment and agents.

QUESTION 196 Which factor distinguishes a lattice network from a random network?

REPEATED IN 2 FILES

- A. Lattice networks have a fixed structure.
- B. Random networks have higher clustering coefficients.
- C. Lattice networks use preferential attachment.
- D. Random networks are infinite.

CORRECT ANSWER A. Lattice networks have a fixed structure.

QUESTION 197 What does "parameter sweep" involve in simulation modeling?

REPEATED IN 2 FILES

- A. Testing random inputs for errors.
- B. Exhaustively varying input values to study model behavior.
- C. Randomizing agent movements in the environment.
- D. Running a model once with optimal parameters.

CORRECT ANSWER B. Exhaustively varying input values to study model behavior.

QUESTION 198 What is the main purpose of using a scale-free network in ABMs?

REPEATED IN 2 FILES

- A. To ensure equal connectivity among nodes
- B. To replicate real-world networks where some nodes have many connections
- C. To minimize clustering coefficients
- D. To simulate a fully disconnected graph

CORRECT ANSWER B. To replicate real-world networks where some nodes have many connections

QUESTION 199 In NetLogo, what is the purpose of the "tick" command?

REPEATED IN 2 FILES

- A. To define new agents
- B. To create patches
- C. To advance the simulation by one step
- D. To generate random numbers

CORRECT ANSWER C. To advance the simulation by one step

QUESTION 200 What type of data is best visualized using histograms in ABMs?

REPEATED IN 2 FILES

- A. Agent movement paths
- B. Distribution of agent attributes
- C. Network topology
- D. Environmental gradients

CORRECT ANSWER B. Distribution of agent attributes

QUESTION 201 What is the primary role of an agent in an agent-based model?

REPEATED IN 2 FILES

- A. To define the simulation boundaries
- B. To execute a predefined sequence of events
- C. To interact with the environment and other agents
- D. To generate random outcomes

CORRECT ANSWER C. To interact with the environment and other agents

QUESTION 202 Which feature of ABMs allows for emergent behavior to be studied?

REPEATED IN 2 FILES

- A. Centralized control of agents
- B. Random number generation
- C. Local interactions between agents
- D. Predefined global outcomes

CORRECT ANSWER C. Local interactions between agents

QUESTION 203 Which NetLogo feature allows users to visualize agent movement paths?

REPEATED IN 2 FILES

- A. Plot widget
- B. View screen
- C. Monitor widget
- D. GIS extension

CORRECT ANSWER B. View screen

QUESTION 204 In simulation modeling, what does "stochastic" refer to?

REPEATED IN 2 FILES

- A. Deterministic outcomes
- B. Outcomes influenced by randomness
- C. Predefined sequences of events
- D. Absence of agent interactions

CORRECT ANSWER B. Outcomes influenced by randomness

QUESTION 205 What is the primary function of "turtles" in NetLogo?**REPEATED IN 2 FILES**

- A. Represent agents that can move and act
- B. Define static environmental properties
- C. Visualize network connections
- D. Generate statistical summaries

CORRECT ANSWER A. Represent agents that can move and act**QUESTION 206 What is a typical application of ABMs in economics?****REPEATED IN 2 FILES**

- A. Simulating predator-prey interactions
- B. Modeling market dynamics and consumer behavior
- C. Testing spatial layouts of cities
- D. Exploring environmental diffusion

CORRECT ANSWER B. Modeling market dynamics and consumer behavior**QUESTION 207 In NetLogo, what does "inspect" allow users to do?****REPEATED IN 2 FILES**

- A. Visualize patch distributions
- B. Change the properties of individual agents
- C. Edit the environment layout
- D. Automate simulations

CORRECT ANSWER B. Change the properties of individual agents**QUESTION 208 What is the primary benefit of using GIS extensions in NetLogo?****REPEATED IN 2 FILES**

- A. To simulate agent interactions in 3D environments
- B. To integrate geographic data into simulations
- C. To visualize network connections in real time
- D. To automate simulation runs

CORRECT ANSWER B. To integrate geographic data into simulations

QUESTION 209 What type of behavior is commonly studied using flocking models in ABMs?

REPEATED IN 2 FILES

- A. Competitive market strategies
- B. Collective movement of agents
- C. Predator-prey interactions
- D. Disease spread

CORRECT ANSWER B. Collective movement of agents

QUESTION 210 In agent-based models, what is "agent heterogeneity"?

REPEATED IN 2 FILES

- A. All agents having the same behavior
- B. Agents having diverse attributes and behaviors
- C. Agents moving in predefined patterns
- D. Agents always following global rules

CORRECT ANSWER B. Agents having diverse attributes and behaviors

QUESTION 211 Which tool in NetLogo is used to track changes in variables over time?

REPEATED IN 2 FILES

- A. View screen
- B. BehaviorSpace
- C. Plot widget
- D. Monitor widget

CORRECT ANSWER C. Plot widget

QUESTION 212 What does "parameter sensitivity analysis" involve in ABMs?

REPEATED IN 2 FILES

- A. Testing the randomness of outcomes
- B. Exploring how changes in parameters affect results
- C. Comparing model outputs to real-world data
- D. Optimizing simulation speed

CORRECT ANSWER B. Exploring how changes in parameters affect results

QUESTION 213 What does "toroidal topology" ensure in agent-based simulations?

REPEATED IN 2 FILES

- A. Infinite agent interactions
- B. Seamless movement across boundaries
- C. Centralized control of agents
- D. Finite boundaries with no agent interaction

CORRECT ANSWER B. Seamless movement across boundaries

QUESTION 214 Which factor is most likely to influence the outcome of an ABM simulation?

REPEATED IN 2 FILES

- A. Random initialization of agents
- B. Fixed initial states of all agents
- C. Absence of stochastic elements
- D. No variation in parameters

CORRECT ANSWER A. Random initialization of agents

QUESTION 215 What is the primary use of the "export-world" command in NetLogo?

REPEATED IN 2 FILES

- A. To save all agent and environment data
- B. To export a graphical snapshot of the simulation
- C. To reset the simulation world
- D. To automate multiple model runs

CORRECT ANSWER A. To save all agent and environment data

QUESTION 216 Which type of network is often used to model disease spread in ABMs?

REPEATED IN 2 FILES

- A. Scale-free networks
- B. Small-world networks
- C. Random networks
- D. Hierarchical networks

CORRECT ANSWER B. Small-world networks

QUESTION 217 In the context of ABMs, what is "adaptive behavior"?

REPEATED IN 2 FILES

- A. Agents always following predefined rules
- B. Agents changing their behavior based on the environment
- C. Agents interacting with other agents in random ways
- D. Static decision-making by agents

CORRECT ANSWER B. Agents changing their behavior based on the environment

QUESTION 218 What does the "observer" in NetLogo primarily control?

REPEATED IN 2 FILES

- A. The properties of individual agents
- B. The visualization of plots and monitors
- C. The initialization and execution of the simulation
- D. Random movements of agents

CORRECT ANSWER C. The initialization and execution of the simulation

QUESTION 219 In an agent-based model, which of the following represents emergent behavior?

REPEATED IN 2 FILES

- A. Predefined patterns followed by all agents
- B. Global patterns arising from local interactions
- C. Random movement of agents in a grid
- D. Fixed outcomes based on agent initialization

CORRECT ANSWER B. Global patterns arising from local interactions

QUESTION 220 What is the primary advantage of using hexagonal grids over square grids in ABMs?

REPEATED IN 2 FILES

- A. Simplified agent programming
- B. Reduced number of neighbors
- C. Better symmetry for distance calculations
- D. Easier visualization of agent behavior

CORRECT ANSWER C. Better symmetry for distance calculations

QUESTION 221 Which ABM model is commonly used to study segregation dynamics?

REPEATED IN 2 FILES

- A. Predator-Prey Model
- B. Schelling's Model
- C. Game Theory Model
- D. Flocking Model

CORRECT ANSWER B. Schelling's Model

QUESTION 222 What is a common limitation of agent-based modeling?

REPEATED IN 2 FILES

- A. Inability to include stochastic elements
- B. Difficulty in scaling simulations with many agents
- C. Restriction to static environments
- D. Lack of local interactions between agents

CORRECT ANSWER B. Difficulty in scaling simulations with many agents

QUESTION 223 Which characteristic is typical of emergent phenomena in ABMs?

REPEATED IN 2 FILES

- A. Pre-programmed global behavior
- B. Patterns resulting from agent interactions
- C. Agents operating without rules
- D. Behavior unrelated to initial conditions

CORRECT ANSWER B. Patterns resulting from agent interactions

QUESTION 224 Which NetLogo command is used to reset all agents and environment variables to their initial state?

REPEATED IN 2 FILES

- A. reset-ticks
- B. clear-all
- C. setup
- D. initialize-world

CORRECT ANSWER C. setup

QUESTION 225 What is a key feature of the Game of Life ABM?**REPEATED IN 2 FILES**

- A. Stochastic agent interactions
- B. Continuous agent movement
- C. Simple rules leading to complex patterns
- D. Hierarchical decision-making among agents

CORRECT ANSWER C. Simple rules leading to complex patterns**QUESTION 226 What does the "patches-own" keyword define in NetLogo?****REPEATED IN 2 FILES**

- A. Variables specific to agents
- B. Variables specific to the environment grid
- C. Global variables for all agents
- D. Shared variables between patches and turtles

CORRECT ANSWER B. Variables specific to the environment grid**QUESTION 227 In NetLogo, what is the purpose of the "go" procedure?****REPEATED IN 2 FILES**

- A. To define initial parameters
- B. To execute simulation steps repeatedly
- C. To reset the environment
- D. To export model data

CORRECT ANSWER B. To execute simulation steps repeatedly**QUESTION 228 How does "agent heterogeneity" enhance the realism of ABMs?****REPEATED IN 2 FILES**

- A. By enforcing uniform agent rules
- B. By introducing diversity in agent attributes and behaviors
- C. By removing randomness from simulations
- D. By limiting interactions between agents

CORRECT ANSWER B. By introducing diversity in agent attributes and behaviors

QUESTION 229 Which NetLogo feature is used to analyze trends in variables during a simulation?

REPEATED IN 2 FILES

- A. BehaviorSpace
- B. Plot widget
- C. Monitor widget
- D. Command Center

CORRECT ANSWER B. Plot widget

QUESTION 230 In ABMs, what does "scalability" refer to?

REPEATED IN 2 FILES

- A. The ability to execute simulations faster
- B. The capacity to handle an increasing number of agents
- C. The complexity of agent interactions
- D. The randomness of simulation results

CORRECT ANSWER B. The capacity to handle an increasing number of agents

QUESTION 231 What is the primary role of diffusion in environmental modeling?

REPEATED IN 2 FILES

- A. To model agent decision-making
- B. To represent the spread of substances or information
- C. To simulate agent mobility
- D. To reduce environmental complexity

CORRECT ANSWER B. To represent the spread of substances or information

QUESTION 232 What does the "ticks" primitive in NetLogo track?

REPEATED IN 2 FILES

- A. The number of agents in the simulation
- B. The number of simulation steps elapsed
- C. The current state of all patches
- D. The total number of executed commands

CORRECT ANSWER B. The number of simulation steps elapsed

QUESTION 233 What is the advantage of "parallel processing" in large-scale ABMs?

REPEATED IN 2 FILES

- A. It simplifies the agent behavior rules.
- B. It reduces computation time by distributing tasks.
- C. It eliminates the need for parameter sweeps.
- D. It minimizes agent interactions.

CORRECT ANSWER B. It reduces computation time by distributing tasks.

QUESTION 234 Which feature of ABMs allows for local rules to lead to global patterns?

REPEATED IN 2 FILES

- A. Emergent behavior
- B. Centralized control
- C. Deterministic updates
- D. Predefined global states

CORRECT ANSWER A. Emergent behavior

QUESTION 235 In NetLogo, what does "patches-own" allow you to do?

REPEATED IN 2 FILES

- A. Assign properties specific to each agent
- B. Define variables unique to grid cells
- C. Share variables between patches and turtles
- D. Randomize agent positions

CORRECT ANSWER B. Define variables unique to grid cells

QUESTION 236 What is a common metric used to evaluate network robustness in ABMs?

REPEATED IN 2 FILES

- A. Clustering coefficient
- B. Degree distribution
- C. Average path length
- D. Network connectivity

CORRECT ANSWER D. Network connectivity

QUESTION 237 Which approach is typically used to increase the scalability of ABMs?

REPEATED IN 2 FILES

- A. Simplifying agent interactions
- B. Reducing the number of agents
- C. Using cloud computing resources
- D. Eliminating environmental factors

CORRECT ANSWER C. Using cloud computing resources

QUESTION 238 What is "stochasticity" in agent-based modeling?

REPEATED IN 2 FILES

- A. Predefined outcomes for all agents
- B. Randomness introduced into agent behavior
- C. Absence of interaction between agents
- D. Centralized control over agent actions

CORRECT ANSWER B. Randomness introduced into agent behavior

QUESTION 239 Which type of topology is best for modeling localized agent interactions?

REPEATED IN 2 FILES

- A. Infinite plane topology
- B. Bounded topology
- C. Toroidal topology
- D. Network topology

CORRECT ANSWER B. Bounded topology

QUESTION 240 How does "sensitivity analysis" benefit model validation?

REPEATED IN 2 FILES

- A. By reducing the number of variables in the model
- B. By measuring the impact of parameter changes on outcomes
- C. By ensuring agents follow identical rules
- D. By eliminating randomness in simulations

CORRECT ANSWER B. By measuring the impact of parameter changes on outcomes

QUESTION 241 In ABMs, what does the term "heterogeneous agents" refer to?

REPEATED IN 2 FILES

- A. Agents with identical properties and behaviors
- B. Agents with varied properties and behaviors
- C. Agents that do not interact with their environment
- D. Agents that follow centralized rules

CORRECT ANSWER B. Agents with varied properties and behaviors

QUESTION 242 What is the primary purpose of a "global variable" in NetLogo?

REPEATED IN 2 FILES

- A. To store values shared by all agents
- B. To define agent-specific behaviors
- C. To create subgroups of agents
- D. To initialize the environment grid

CORRECT ANSWER A. To store values shared by all agents

QUESTION 243 An _____ is defined as an instantaneous occurrence that might change the state of the system.

- A. Occasion
- B. State
- C. Event
- D. Moment

CORRECT ANSWER C. Event

QUESTION 244 In the three-phase approach, events are considered to be activities of duration _____ units.

- A. No-time
- B. Zero-time
- C. Milliseconds
- D. Few seconds

CORRECT ANSWER B. Zero-time

QUESTION 245 The definition of the model _____ provides a static description of the model.

- A. Entities
- B. Components
- C. Variables
- D. Parameters

CORRECT ANSWER B. Components

QUESTION 246 The solution usually consists of one or more numerical parameters, which are called:

- A. Measure of performance
- B. Simulation-generated data
- C. Mathematical parameter
- D. Simulation Model

CORRECT ANSWER A. Measure of performance

QUESTION 247 In _____ to an activity, a delay's duration is not specified by the modeler ahead of time.

- A. Order
- B. Contrast
- C. Comparison
- D. Sequence

CORRECT ANSWER B. Contrast

QUESTION 248 If managers have unreasonable expectations, if they ask for too much too soon, or if the power of simulation is overestimated, simulation might not be:

- A. Reliable
- B. Appropriate
- C. Suitable
- D. Suggested

CORRECT ANSWER B. Appropriate

QUESTION 249 Simulation can be performed faster today than yesterday and will be even faster tomorrow, because of advances in _____.

- A. Software
- B. Methods
- C. Technology
- D. Hardware

CORRECT ANSWER D. Hardware

QUESTION 250 Time can be compressed or expanded to allow for a _____ of the phenomena under _____.

- A. Speed-up or slow-down, Investigation
- B. Speed-up, Observation
- C. Slow-down, Analysis
- D. Both a & b

CORRECT ANSWER A. Speed-up or slow-down, Investigation

QUESTION 251 Vendors of _____ software have been actively developing packages that contain models that need only _____ data for their operation.

- A. Simulation, input
- B. Compiled, numerical
- C. Sampled, theoretical
- D. Analyzed, hypothetical

CORRECT ANSWER A. Simulation, input

QUESTION 252 The properties of a given entity are called:

- A. Model
- B. List
- C. Attributes
- D. Entity

CORRECT ANSWER C. Attributes

QUESTION 253 A _____ is defined as a group of objects that are joined together in some _____ interaction or interdependence:

- A. System, regular
- B. Application, irregular
- C. System, dynamic
- D. System, random

CORRECT ANSWER A. System, regular

QUESTION 254 The sequence of actions that a simulator must perform to advance the clock and build a new system snapshot is called the _____.

- A. Event-scheduling/time-advance algorithm
- B. Frequency / times algorithm
- C. Time-managing / Advanced algorithm
- D. Time-Scheduling / advance algorithm

CORRECT ANSWER A. Event-scheduling/time-advance algorithm

QUESTION 255 The _____ gained during the designing of a _____ could be of great value toward suggesting improvement in the system under investigation:

- A. Screening, Estimation
- B. Analysis, Statistics
- C. Observation, Calculations
- D. Knowledge, Simulation Model

CORRECT ANSWER D. Knowledge, Simulation Model

QUESTION 256 Many simulation software vendors have developed _____ capabilities within their packages for performing very thorough analysis.

- A. None
- B. Inference based analysis
- C. Input-analysis
- D. Output-analysis

CORRECT ANSWER D. Output-analysis

QUESTION 257 If no data is available, not even estimates, simulation _____.

- A. Is most suitable
- B. None
- C. Is feasible
- D. Is not advised

CORRECT ANSWER D. Is not advised

QUESTION 258 A duration of time of unspecified indefinite length, which is _____ until it ends.

- A. Fixed
- B. Changes
- C. Specified
- D. Not known

CORRECT ANSWER D. Not known

QUESTION 259 Closed-form models are not able to analyze most of the _____ that are encountered in practice.

- A. Simple system
- B. Complex systems
- C. Complicated system
- D. Natural system

CORRECT ANSWER B. Complex systems

QUESTION 260 A system composed of significant life events that can affect adaptation.

- A. Macrosystem
- B. Adaptive system
- C. Microsystem
- D. Chronosystem

CORRECT ANSWER D. Chronosystem

QUESTION 261 The mechanism for advancing simulation time and guaranteeing that all events occur in correct chronological order is based on the _____.

- A. Past event list (PEL)
- B. Future event list (FEL)
- C. Present event list (PEL)
- D. Future Indefinite event list (FIEL)

CORRECT ANSWER B. Future event list (FEL)

QUESTION 262 In many years of _____, not one problem was encountered that could have been solved by a closed-form solution.

- A. Consulting practice
- B. Conclusions
- C. Observation
- D. Analysis

CORRECT ANSWER A. Consulting practice

QUESTION 263 Circular interactions that systems engage in such that they influence one another:

- A. Efficient Transaction
- B. Adaptation
- C. Homeostasis
- D. Reciprocal Transactions

CORRECT ANSWER D. Reciprocal Transactions

QUESTION 264 Rate of energy transfer between the system and its environment during the time it is functioning:

- A. Frequency
- B. Efficiency
- C. None
- D. Throughput

CORRECT ANSWER D. Throughput

QUESTION 265 A runtime and downtime are examples of activities, and _____ and _____ are primary events.

- A. End of debugging, end of the downtime
- B. End of compilation, end of uptime
- C. End of runtime, end of the downtime
- D. End of runtime, end of uptime

CORRECT ANSWER C. End of runtime, end of the downtime

QUESTION 266 More precisely, a process is the life cycle of _____ entity.

- A. Two
- B. Three
- C. One
- D. Four

CORRECT ANSWER C. One

QUESTION 267 A simulation is the imitation of the operation of a real-world process or system over time

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- A. Imaginary word process
- B. Real-world process
- C. Both A & B
- D. Neither A nor B

CORRECT ANSWER B. Real-world process

QUESTION 268 Cognitive Agent-based Computing Framework by Niazi andpresents detailed instructions on how to build Agent-based Models

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- A. ahmad
- B. rehman
- C. hussain
- D. wilensky

CORRECT ANSWER C. hussain

QUESTION 269 _____ is a numerical computational technique used in conjunction with dynamic mathematical models.

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- A. Analysis
- B. System Simulation
- C. Dynamic Computation
- D. All of the above

CORRECT ANSWER B. System Simulation

QUESTION 270 Which model follows the changes over time that results from the system activities.

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- A. Dynamic Model
- B. Static Model
- C. Analytical Model
- D. All of the above

CORRECT ANSWER A. Dynamic Model

QUESTION 271 Which model in system modeling depicts the dynamic behavior of the system?

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- A. Context Model
- B. Behavioral Model
- C. Data Model
- D. All of the above

CORRECT ANSWER B. Behavioral Model

QUESTION 272 Which model in system modeling depicts the static nature of the system?

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- A. Behavioral Model
- B. Context Model
- C. Data Model
- D. Structural Model

CORRECT ANSWER D. Structural Model

QUESTION 273 _____ can sometimes present new phenomena that are not necessarily about some real-world phenomenon but are thought experiments on possible computations.

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- A. Agent
- B. Event
- C. Model
- D. Activity

CORRECT ANSWER C. Model

QUESTION 274 Emergent phenomena can be described as existing on at least _____ levels.

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- A. Two
- B. Three
- C. Four
- D. Five

CORRECT ANSWER A. Two

QUESTION 275 _____ is a collection of entities that interact together over time to accomplish one or more goals.

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- A. System
- B. Model
- C. Simulation
- D. Activities

CORRECT ANSWER A. System

QUESTION 276 NetLogo calls its prototypical agent a _____.

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- A. Turtle
- B. Patches
- C. Links
- D. Observer

CORRECT ANSWER A. Turtle

QUESTION 277 _____ is define as a system that are composed of multiple individual elements that interact with each other.

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- A. Complex System
- B. Agent-based System
- C. Cognitive System
- D. Queueing System

CORRECT ANSWER A. Complex System

QUESTION 278 _____ network in which each node is randomly connected to other individual node.

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- A. Lattice
- B. Random
- C. Scale-free
- D. Small-world

CORRECT ANSWER B. Random

QUESTION 279 Model accuracy is evaluated through _____.

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- A. Validation and Verification
- B. Verification and Replication
- C. Replication and Validation
- D. Calibration and Validation

CORRECT ANSWER A. Validation and Verification

QUESTION 280 _____ levels of agent-based modeling is present in CABC framework.

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- A. Two
- B. Three
- C. Four
- D. Five

CORRECT ANSWER B. Three

QUESTION 281 The linear congruential method produces the sequence of integers between _____.

★ **MOST IMPORTANT - PAST PAPER**

- A. 0 and 1
- B. 0 and $m-1$
- C. m and 1
- D. 0 and m

CORRECT ANSWER B. 0 and $m-1$

QUESTION 282 _____ distribution operates on acceptance rejection technique.

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- A. Binomial distribution
- B. Triangular distribution
- C. Exponential distribution
- D. Normal distribution

CORRECT ANSWER A. Binomial distribution

QUESTION 283 An entity of interest in our system is describe by an/a _____.

★ **MOST IMPORTANT - PAST PAPER**

- A. Activity
- B. Event
- C. Agent
- D. Attribute

CORRECT ANSWER B. Event

QUESTION 284 _____ is defined as a relationship between independent variable x and a dependent variable y .

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- A. Linear Correlation
- B. Linear Regression
- C. Quadractic Regression
- D. Nonlinear Correlation

CORRECT ANSWER B. Linear Regression

QUESTION 285 _____ can connect a device from anywhere.

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- A. CAN
- B. VAN
- C. SAN
- D. LAN

CORRECT ANSWER A. CAN

QUESTION 286 _____ distribution was created on the physical basis.

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- A. Beta
- B. Gamma
- C. Binomial
- D. Weibull

CORRECT ANSWER C. Binomial

QUESTION 287 _____ equation used in triangular distribution

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- A. $a \leq b \leq c$
- B. $a \leq c \leq b$
- C. $a \leq c \geq b$
- D. $b \leq c \leq a$

CORRECT ANSWER B. $a \leq c \leq b$

QUESTION 288 Deterministic system that is difficult to predict is known as _____

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- A. Discrete
- B. Adaptive
- C. Chaotic
- D. Stochastic

CORRECT ANSWER C. Chaotic

QUESTION 289 _____ is the of finding the parameters and initial conditions that cause the model to match with real world data.

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- A. Calibration
- B. Replication
- C. Validation
- D. Verification

CORRECT ANSWER A. Calibration

QUESTION 290 The major goal of _____ is to focus on explaining and summarizing a specific dataset.

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- A. probability
- B. Mathematics
- C. Descriptive Statistics
- D. All of the above

CORRECT ANSWER C. Descriptive Statistics

QUESTION 291 A precise statement of components to be used and the relationships among them

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- A. Formal Model
- B. Informal Model
- C. Simulated Model
- D. Conceptual Model

CORRECT ANSWER A. Formal Model

QUESTION 292 If some of the detected faults are not removed with certainty or new faults introduced during debugging process then it is called _____.

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- A. Incomplete perfect Debugging
- B. Perfect Debugging
- C. Imperfect Debugging
- D. All of the above

CORRECT ANSWER C. Imperfect Debugging

QUESTION 293 In modelling and simulations, useful statistical models have

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- A. Inventory and maintainability
- B. Limited data
- C. Queueing system
- D. All of these

CORRECT ANSWER D. All of these

QUESTION 294 A _____ is the model of an application that the designers want users to understand.

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- A. conceptual model
- B. Computational Model
- C. Dynamic Model
- D. Formal Model

CORRECT ANSWER A. conceptual model

QUESTION 295 If two-sample-t confidence interval approach is used. The point estimate of the ____ difference is:

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- A. Mean
- B. Average
- C. Variance
- D. Standard deviation

CORRECT ANSWER A. Mean

QUESTION 296 The _____ of the customers in a queue and determining that which customer should come for service when the server is free

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- A. Logical ordering
- B. Customized ordering
- C. Simulated ordering
- D. Random ordering

CORRECT ANSWER A. Logical ordering

QUESTION 297 An original model and replicated model can differ across at least six dimensions. Which of the following is not part of this six dimensions

I. Time

II. Hardware

III. Languages

IV. Toolkits

V. Algorithms

VI. Authors

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- A. I
- B. II, III, VI
- C. VI, V
- D. III, IV

CORRECT ANSWER A. I

QUESTION 298 When you are trying to collect statistical results from an ABM you should run the model _____ and collect different results at different points

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- A. Multiple times
- B. Once
- C. Two times
- D. Three times

CORRECT ANSWER A. Multiple times

QUESTION 299 What is the primary difference between finite and infinite population models in queuing theory?

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- A. Service rate
- B. Arrival rate
- C. Queue length
- D. Waiting time

CORRECT ANSWER B. Arrival rate

QUESTION 300 What are the characteristics of the Linear Congruential Method (LCM) for generating random numbers?

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- A. Produces sequences with shorter periods, yields real numbers, and is rarely used.
- B. Yields sequences with longer periods, produces the sequence of integers, and is widely used.
- C. Produces sequences with longer periods, yields real numbers, and is moderately used.
- D. Yields sequences with shorter periods, produces the sequence of integers, and is rarely used.

CORRECT ANSWER B. Yields sequences with longer periods, produces the sequence of integers, and is widely used.